Lecture 14 - L'Hôpital's Rule

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Updates

We have Exam 2 next week!
Practice exams and the review sheets are both out on Canvas, compiled lecture will be uploaded after class.
Solutions to the practice exam and review sheet will be uploaded Thursday night to give you all time to try out the problems before looking at the solutions.
If you're feeling worried about the exam, please feel free to come to my office hours or email me to set up a time to meet :)
Re: quizzes. I'll grind out quiz grading tonight and tomorrow. Let me know if you want me to give feedback on a specific quiz of yours earlier though :)

Quiz Time

Let’s get into neat rows, spread out, and slay this quiz :)

L’Hôpital’s Rule

Limit Review I

Recall from the first two weeks of class:
we use *limits* to determine the values that functions approximate as their inputs approximate a given value.

They were very useful in examining *indeterminate forms*: expressions like $\frac{f(x)}{g(x)}$ around values $x = x_0$ where $f(x_0) = g(x_0) = 0$ or ∞.
Our tools included:

- Factoring out $(x - x_0)$
- Replacing the limit with the limit of a continuous function
- Addition/subtraction/product/quotient computations
- Squeeze theorem
- Etc.

Limit Review II

We also defined derivatives in terms of limits:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{1}$$

But then quickly pivoted to constructing and working with the *synthetic* derivative rules, e.g.:

- Power Rule: $\frac{d}{dx} x^n = n x^{n-1}$
- Exponential/Logarithm rule: $\frac{d}{dx} e^x = e^x$, $\frac{d}{dx} \ln(x) = \frac{1}{x}$
- Trig Rules: $\frac{d}{dx} \sin(x) = \cos(x)$
- Sum/Difference rule:

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$$

- Product Rule: $\frac{d}{dx} f(x)g(x) = \left(\frac{d}{dx} f(x)\right) \cdot g(x) + f(x) \cdot \left(\frac{d}{dx} g(x)\right)$
- Chain Rule: $\frac{d}{dx} g(f(x)) = \frac{dg}{du}\Big|_{u=f(x)} \cdot \frac{df}{dx}$.

Derivative Review I

The importance of the derivative came from it's interpretation as *the best linear approximation*:

$$\lim_{h \to 0} \frac{|f(x) - (f(x_0) + f'(x_0)(x - x_0))|}{h} = 0 \tag{2}$$

or, in practical terms:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + O((x - x_0)^2) \tag{3}$$

We also saw the *quadratic approximation*:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + O((x - x_0)^3) \tag{4}$$

L'Hôpital's Rule I

Because of our ability to approximate functions via their linear/quadratic approximations, we gain another tool to put in our limit computing toolkit when we encounter indeterminate forms.

If $f(x)$ and $g(x)$ are two functions that are:

1. 0 when $x = x_0$
2. Differentiable at $x = x_0$

then, assuming $g'(x_0) \neq 0$, an application of squeeze theorem gives us:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f(x_0) + f'(x_0)(x - x_0)}{g(x_0) + g'(x_0)(x - x_0)} = \lim_{x \to x_0} \frac{f'(x_0)(x - x_0)}{g'(x_0)(x - x_0)} = \lim_{x \to x_0} \frac{f'(x_0)}{g'(x_0)} \tag{5}$$

We can also work around $g'(x_0)$ being zero by instead reducing

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \tag{6}$$

L'Hôpital's Rule II

(Mini) L'Hôpital's Rule

If $f(x)$, $g(x)$ are two functions that are:

1. Zero at $x = x_0$ (i.e. $f(x_0) = g(x_0) = 0$)
2. Differentiable at $x = x_0$

then:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \tag{7}$$

In other words, we can compute limits of indeterminate forms by *differentiating the two functions* to kill off any obstructions to continuity.

L'Hôpital's Rule III

This comes from a generalization of both linear and quadratic approximation, arguably the most important concept taught in Calculus II:

Taylor's Theorem

For $f(x)$ a function whose higher derivatives $\frac{d^n f}{dx^n}\,|_{x=x_0}$ all exist one can form the *Taylor Polynomial* of $f(x)$ at $x = x_0$ defined as:

$$T_f(x) := \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \tag{8}$$

and as $x \to x_0$ the error term $E(x) = |f(x) - T_f(x)|$ approaches zero *extremely rapidly*. (And in most nice cases it turns out that $f(x) = T_f(x)$ *literally* on a small interval containing x_0).

(this is not material you are expected to remember for this class)

L'Hôpital's Rule IV

L'Hôpital's Rule is then just what happens when you compute $\lim_{x \to x_0} \frac{f(x)}{g(x)}$ by replacing it with $\lim_{x \to x_0} \frac{T_f(x)}{T_g(x)}$ using the Squeeze Theorem. E.g.:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f'''(x_0)}{6}(x - x_0)^3 + \ldots}{g(x_0) + g'(x_0)(x - x_0) + \frac{g''(x_0)}{2}(x - x_0)^2 + \frac{g'''(x_0)}{6}(x - x_0)^3 + \ldots} \tag{9}$$

$$= \lim_{x \to x_0} \frac{f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f'''(x_0)}{6}(x - x_0)^3 + \ldots}{g'(x_0)(x - x_0) + \frac{g''(x_0)}{2}(x - x_0)^2 + \frac{g'''(x_0)}{6}(x - x_0)^3 + \ldots} \tag{10}$$

$$= \lim_{x \to x_0} \frac{f'(x_0) + \frac{f''(x_0)}{2}(x - x_0) + \frac{f'''(x_0)}{6}(x - x_0)^2 + \ldots}{g'(x_0) + \frac{g''(x_0)}{2}(x - x_0) + \frac{g'''(x_0)}{6}(x - x_0)^2 + \ldots} \tag{11}$$

L'Hôpital's Rule V

What if $\lim_{x \to x_0} \frac{f'(x)}{g'(x)}$ is another $\frac{0}{0}$ indeterminate form? e.g. $f'(x_0) = g'(x_0) = 0$?

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x_0) + \frac{f''(x_0)}{2}(x - x_0) + \frac{f'''(x_0)}{6}(x - x_0)^2 + \ldots}{g'(x_0) + \frac{g''(x_0)}{2}(x - x_0) + \frac{g'''(x_0)}{6}(x - x_0)^2 + \ldots} \tag{12}$$

$$= \lim_{x \to x_0} \frac{\frac{f''(x_0)}{2}(x - x_0) + \frac{f'''(x_0)}{6}(x - x_0)^2 + \ldots}{\frac{g''(x_0)}{2}(x - x_0) + \frac{g'''(x_0)}{6}(x - x_0)^2 + \ldots} \tag{13}$$

$$= \lim_{x \to x_0} f''(x_0) + \frac{f'''(x_0)}{3}(x - x_0) + \ldots g''(x_0) + \frac{g'''(x_0)}{3}(x - x_0) + \ldots \tag{14}$$

In other words, we just L'Hôpital it again!

L'Hôpital's Rule VI

This gives us a general procedure to compute limits of the form $\lim_{x \to x_0} \frac{f(x)}{g(x)}$:

L'Hôpital's Process for Computing Limits of the Indeterminate Form $\frac{0}{0}$

To compute $\lim_{x \to x_0} \frac{f(x)}{g(x)}$:

1. Compute $f(x_0)$ and $g(x_0)$.
2. If $f(x_0) \neq 0$ and $g(x_0) \neq 0$, then $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{f(x_0)}{g(x_0)}$.
3. If $f(x_0) \neq 0$ but $g(x_0) = 0$ then we're ending up with something like $\frac{N}{0}$ so: $\lim_{x \to x_0} \frac{f(x)}{g(x)}$ DNE
4. If $f(x_0) = g(x_0) = 0$ and $f(x)$, $g(x)$ are differentiable at $x = x_0$, replace $f(x)$ with $f'(x)$ and $g(x)$ with $g'(x)$ and restart step 1.

L'Hôpital's Rule VII

What about other indeterminate forms like:

$$\frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad 0^0, \quad \infty^0, \quad 1^\infty, \quad \infty - \infty?$$

If L'Hôpital's rule helps us compute limits of indeterminate forms like $\frac{0}{0}$ why not just try and force those above forms into something related enough and that looks like $\frac{0}{0}$!

E.g. if we had $\lim_{x \to 0} \frac{\cot(x)}{\ln(x^2)}$ we see this is of the form $\frac{\infty}{\infty}$ but we can change it to:

$$\lim_{x \to 0} \frac{\cot(x)}{\ln(x^2)} = \left(\lim_{x \to 0} \frac{\frac{1}{\ln(x^2)}}{\frac{1}{\cot(x)}} \right)$$

which is of the form $\frac{0}{0}$.

L'Hôpital's Rule Examples I

We can then compute:

$$\lim_{x\to 0}\frac{\cot(x)}{\ln(x^2)} = \lim_{x\to 0}\frac{\frac{1}{\ln(x^2)}}{\tan(x)} \tag{15}$$

$$= \lim_{x\to 0}\frac{\frac{d}{dx}(\ln(x^2))^{-1}}{\frac{d}{dx}\tan(x)} \tag{16}$$

$$= \lim_{x\to 0}\frac{2x\cdot\frac{1}{(x^2)}\cdot -1\cdot(\ln(x^2))^{-2}}{\sec(x)^2} \tag{17}$$

$$= \lim_{x\to 0}\frac{\frac{-2\ln(x^2)}{x}}{\sec(x)^2} \tag{18}$$

since $\sec(0)^2 = 1$, and since the numerator is an indeterminate form of type $\frac{\infty}{0}$, we can conclude this limit DNE.

L'Hôpital's Rule Examples II

If you're a fan of Mean Girls, you might be interested in computing:

$$\lim_{x \to 0} \frac{\ln(1-x) - \sin(x)}{1 - \cos(x)^2} \tag{19}$$

We check @ $x = 0$, we'd get: $\frac{\ln(1-0) - \sin(0)}{1 - \cos(0)^2} = \frac{\ln(1) - 0}{1 - 1^2} = \frac{0}{0}$ so this is already in a form amenable to L'Hôpital's Rule!

$$\lim_{x \to 0} \frac{\ln(1-x) - \sin(x)}{1 - \cos(x)^2} = \lim_{x \to 0} \frac{\frac{d}{dx}(\ln(1-x) - \sin(x))}{\frac{d}{dx}(1 - \cos(x)^2)} \tag{20}$$

$$= \lim_{x \to 0} \frac{\frac{-1}{1-x} - \cos(x)}{-2\cos(x) \cdot -\sin(x)} \tag{21}$$

Which at 0 is $\frac{\frac{-1}{1-0} - \cos(0)}{2\cos(0)\sin(0)} = \frac{-2}{0}$ so the limit does not exist!

L'Hôpital's Rule Examples III

Another one:

$$\lim_{x \to 0} x^x \tag{22}$$

Indeterminate form of type 0^0. What can we use to turn exponentiation into a product/quotient of things? Logarithms!

$$\ln\left(\lim_{x \to 0} x^x\right) = \lim_{x \to 0} \ln(x^x) = \lim_{x \to 0} x \ln(x) \tag{23}$$

$$= \lim_{x \to 0} \frac{\ln(x)}{\frac{1}{x}} = \lim_{x \to 0} \frac{\frac{d}{dx}\ln(x)}{\frac{d}{dx}\frac{1}{x}} \tag{24}$$

$$= \lim_{x \to 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \to 0} -\frac{x^2}{x} = 0 \tag{25}$$

So $\ln(\lim_{x \to 0} x^x) = 0$, which means $\lim_{x \to 0} x^x = e^0 = 1$.

L'Hôpital's Rule Cheat Method!

In the previous example we did literal L'Hôpital's rule on an indeterminate form of type $\frac{\infty}{\infty}$. Why is this valid?

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{\frac{1}{g(x)}}{\frac{1}{f(x)}} \tag{26}$$

$$= \lim_{x \to x_0} \frac{\frac{d}{dx}\frac{1}{g(x)}}{\frac{d}{dx}\frac{1}{f(x)}} \tag{27}$$

$$= \lim_{x \to x_0} \frac{\frac{-g'(x)}{g(x)^2}}{\frac{-f'(x)}{f(x)^2}} = \left(\lim_{x \to x_0} \frac{g'(x)}{f'(x)}\right) \cdot \left(\lim_{x \to x_0} \frac{f(x)}{g(x)}\right)^2 \tag{28}$$

from which you can solve that, for $\lim_{x \to x_0} f(x) = \lim_{x \to x_0} g(x) = \infty$

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \tag{29}$$

Superpowered L'Hôpital's Rule

L'Hôpital's Rule

For $f(x)$, $g(x)$ two functions where *either*:

1. $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = 0$ and $f(x), g(x)$ are differentiable at $x = x_0$
2. $\lim_{x\to x_0} |f(x)| = \lim_{x\to x_0} |g(x)| = \infty$ and $\frac{1}{f(x)}, \frac{1}{g(x)}$ are differentiable at $x = x_0$.

Then we have that:

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \lim_{x\to x_0} \frac{f'(x)}{g'(x)} \tag{30}$$

Then to compute the limit of *any* indeterminate form, we need only equate it to some indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and use L'Hôpital's Rule to compute it.

L'Hôpital's Rule Examples IV

Let's consider $\lim_{x\to 0}\frac{\sin(x)}{\cos(x)}$ as well as the following "computation":

$$\lim_{x\to x_0}\frac{\sin(x)}{\cos(x)} = \lim_{x\to x_0}\frac{\frac{d}{dx}\sin(x)}{\frac{d}{dx}\cos(x)} \tag{31}$$

$$= \lim_{x\to 0}\frac{\cos(x)}{-\sin(x)} \tag{32}$$

DNE since $\cos(0) = 1$ while $-\sin(0) = 0$ so we end up with $\frac{1}{0}$ so DNE. Thus $\tan(0)$ does not exist.

WE DIDN'T ACTUALLY CHECK IF $\frac{\sin(x)}{\cos(x)}$ WAS AN INDETERMINATE FORM AT $x = 0$!

$\sin(0) = 0$ and $\cos(0) = 1$ so $\frac{\sin(x)}{\cos(x)}$ is *continuous* at $x = 0$ so

$\lim_{x\to 0}\frac{\sin(x)}{\cos(x)} = \frac{\sin(0)}{\cos(0)} = \frac{0}{1} = 0 = \tan(0)$.

L'Hôpital's Rule Examples V

Finally, let's consider

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x \tag{33}$$

(which actually matters since its the "rate of continuously compounding interest")
What type of indeterminate form? $\lim_{x\to\infty} 1+\frac{1}{x}=1$, so 1^∞.
How do we turn exponentiation into a product/quotient? Log!

$$\ln\left(\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x\right)=\lim_{x\to\infty}\ln\left(\left(1+\frac{1}{x}\right)^x\right) \tag{34}$$

$$=\lim_{x\to\infty} x\ln\left(1+\frac{1}{x}\right)=\lim_{x\to\infty}\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}} \tag{35}$$

$$=\lim_{x\to\infty}\frac{-\frac{1}{x^2}\cdot\frac{1}{1+\frac{1}{x}}}{-\frac{1}{x^2}}=\lim_{x\to\infty}\frac{x}{x+1}=1 \tag{36}$$

So $\ln\left(\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x\right)=1$ and therefore $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x=e$.

Worksheet Time!

Let’s get into groups and work on the worksheet :)

End of Class

That’s all folks :)